25901238 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECECT204 - MULTIMEDIA ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Identify what an octave represents in terms of frequency. [K] / [CO1]

   a). Half the frequency b). Same frequency

   c). Double the frequency d). Triple the frequency

2. Select why impedance matching is important in microphone systems. [U] / [CO2]

   a). To increase echo b). To reduce signal loss and distortion

   c). To increase background noise d). To change the pitch of sound

3. State why crossover improves sound clarity. [U] / [CO3]

   a). Adds noise b). Prevents distortion

   c). Reduces power d). Stores signal

4. Recall the audio file format that uses uncompressed audio data. [K] / [CO1]

a). MP3 b). WAV c). AMR d). AAC

5. Indicate the advantage of fiber optic cable. [U] / [CO2]

a). High bandwidth b). More interference

c). Low speed d). Heavy weight

6. Recall the color used for left channel audio. [K] / [CO1]

a). Red b). White c). Yellow d). Green

7. Recall piezoelectric speaker works on. [K] / [CO2]

a). Light effect b). Magnetism only

c). Piezoelectric effect d). Compression

8. Indicate the importance of rendering. [U] / [CO4]

a). Store power b). Delete file

c). Export final output d). Increase noise

9. Select why GPU memory is important. [U] / [CO2]

a). Handles high-resolution graphics b). Stores sound

c). Increases noise d). Reduces power

10. Indicate why DJs use headphones. [U] / [CO4]

a). Monitor tracks b). Store sound

c). Adjust sound levels d). Encode data

11. Identify the advantage of XLR connectors. [K] / [CO1]

a). Unbalanced signal b). Balanced signal with noise reduction

c). Wireless transmission d). USB storage

12. Indicate the purpose of chroma keying. [U] / [CO2]

a). Increase brightness b). Replace background

c). Compress video d). Amplify sound

13. Recall what a headphone converts. [K] / [CO3]

a). Electrical energy to sound energy b). Sound energy to electrical energy

c). Video signal to audio signal d). Digital signal to analog signal

14. Recognize the difference between OMTP and CTIA standards. [U] / [CO1]

a). Different video resolution

b). Different microphone and ground pin configuration

c). Different file formats

d). Different amplifier classes

15. Select why noise reduction is used. [U] / [CO4]

a). Remove background noise b). Increase noise

c). Store sound d). Compress video

16. Select the video format widely used for mobile devices. [U] / [CO1]

a). AVI b). FLV c). MP4 d). MKV

17. Identify the advantage of horn speaker. [U] / [CO2]

a). High efficiency b). Low volume

c). High noise d). No amplification

18. Identify why feedback occurs in PA systems. [U] / [CO3]

a). Low volume b). Signal loop between mic and speaker

c). No power d). High storage

19. Indicate the platform used for live streaming. [K] / [CO2]

a). Excel b). MS Word c). YouTube Live d). Paint

20. State why effects are used in music. [U] / [CO4]

a). Enhance creativity b). Store files

c). Reduce quality d). Increase noise

21. Interpret the use of SpeakON connectors. [U] / [CO1]

a). For microphone input b). For high-power speaker connections

c). For video streaming d). For USB devices

22. Recognise main component of moving coil speaker. [K] / [CO2]

a). Scanner b). Printer head c). Voice coil d). Router

23. Recall HDMI transmits. [K] / [CO4]

a). Audio only b). Video only

c). Audio and video d). Power only

24. Identify the importance of impedance matching in power amplifiers. [U] / [CO1]

a). To increase file size b). To prevent signal distortion and damage

c). To reduce video resolution d). To store audio

25. Select the advantage of electrodynamic speaker. [U] / [CO2]

a). Wide frequency range b). Low output only

c). High noise d). No magnet

26. Interpret the limitation of RCA connectors. [U] / [CO1]

a). Supports digital signals only b). Unbalanced connection prone to noise

c). High-power transmission d). Fiber-based

27. Select why sensitivity rating matters. [U] / [CO2]

a). Displays text b). Stores sound

c). Compresses video d). Determines loudness output

28. Select the purpose of a subwoofer. [U] / [CO3]

a). Enhance bass b). Reduce noise c). Store sound d). Edit audio

29. State the main function of a pre-amplifier. [K] / [CO1]

a). Reduce noise b). Increase weak signal level

c). Store audio d). Convert analog to digital

30. Indicate why transitions are used. [U] / [CO4]

a). Smooth scene change b). Store data

c). Increase noise d). Reduce size

31. Recall the sound characteristic that helps to distinguish between two sounds of same pitch and loudness. [K] / [CO1]

a). Amplitude b). Frequency c). Timbre d). Intensity

32. Identify the purpose of a directional microphone. [U] / [CO2]

a). Reduce background noise b). Increase storage

c). Compress audio d). Store signal

33. State the purpose of mixer console. [K] / [CO3]

a). Combine audio signals b). Store files

c). Compress video d). Display output

34. Distinguish the characteristic of Class A amplifier. [U] / [CO1]

a). Low distortion and low efficiency b). High distortion and high efficiency

c). No power consumption d). Digital switching only

35. State the function of a video encoder. [K] / [CO2]

a). Converts analog to digital b). Amplifies Signal

c). Compresses video data d). Displays video

[1x35 = 35]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

36. Explain amplitude and frequency of sound with examples. [K] / [CO1]

37. Compare HDMI and DVI. [U] / [CO2]

38. Explain tone control circuit for bass and treble. [U] / [CO3]

39. What is a bitmap image? Mention its advantages and disadvantages. [U] / [CO1]

40. Explain the role of AV tools in movie making. [U] / [CO4]

[5x3 = 15]